Chain rule

1 Given that $y = u^3$ and $u = 2x + 3$, define y in terms of x.

2 Redefine the following functions as composite functions $f(u)$ and $u(x)$, where $u(x)$ is a polynomial:

a $f(x) = \left(5x^3 - 4x^2 + 3x - 5\right)^7$

b $f(x) = \sqrt[4]{2x^2 + 2x + 3}$

c $f(x) = \dfrac{1}{\left(4x^2 - 3x + 5\right)^3}$

3 To differentiate the function $y = \left(2x^4 + 6\right)^5$ using the substitution $u = 2x^4 + 6$:

a Find $\dfrac{du}{dx}$.

b Express y as a function of u.

c Find $\dfrac{dy}{du}$.

d Hence, find $\dfrac{dy}{dx}$.

4 For each of the following:

i Find $\dfrac{dy}{du}$.

ii Find $\dfrac{du}{dx}$.

iii Hence, find $\dfrac{dy}{dx}$.

a $y = (x + 5)^5$, where $y = u^5$ and $u = x + 5$.

b $y = (4x + 3)^{-1}$, where $y = u^{-1}$ and $u = 4x + 3$.

c $y = \sqrt{5 + x^2}$, where $y = \sqrt{u}$ and $u = 5 + x^2$.

d $y = \sqrt[3]{\dfrac{15}{x}}$, where $y = \sqrt[3]{u}$ and $u = \dfrac{15}{x}$.

5 Differentiate $y = (5x - 7)^2$:

a By expanding the brackets first.

b By using the chain rule.

6 Find the derivative of $y = \dfrac{1}{x^2 + 5}$. Give your answer in positive index form.

7 Find the derivative of $y = \sqrt{8x + 5}$ using the chain rule. Give your answer in surd form.

8 Differentiate the following using the chain

a $y=(4x+3)^9$

b $y=\left(2t^7+8t^3+3t+5\right)^{-4}$

c $y=\left(x^2+x^{-3}\right)^3$

d $y=\sqrt[3]{x^2-5x}$

e $y=\left(3x^2-4x+2\right)^4$

f $y=-3\left(3x+4\right)^{10}$

g $y=\dfrac{(9x+7)^{\frac{4}{3}}}{3}$

h $y=\dfrac{1}{(x+6)^5}$

i $y=\dfrac{1}{x^4-4x^3+5x}$

j $y=\dfrac{2}{\sqrt{1+x}}$

k $y=\sqrt[5]{(4x+1)^6}$

l $y=-4\left(\dfrac{1}{3}x+1\right)^{-6}$

m $y=5\sqrt{4-\dfrac{1}{3}x}$

n $y=\dfrac{2}{1-x\sqrt{5}}$

o $y=\left(\sqrt{x}+\dfrac{1}{\sqrt{x}}\right)^8$

p $y=\dfrac{1}{\sqrt[3]{(9-x)^4}}$

q $y=-3\left(x+\dfrac{1}{x}\right)^5$

Gradients, tangents and normals

9 Consider the function $g\left(x\right)=\left(3-x^5\right)^4$. Use the chain rule to evaluate $g'\left(-1\right)$.

10 Consider the function $g\left(x\right)=\sqrt{7-3x^2}$.

a Find $g'\left(1\right)$.

b Is the function increasing or decreasing at $x=1$?

c Find the range of values of x for which the function is increasing or decreasing.

11 Consider the function $y=\sqrt{3-2x}$.

a Find the derivative of the function.

b To find where the tangent would be horizontal, what equation would we need to solve?

c Which of the following x-values has a tangent that is closest to horizontal?

A $-\dfrac{3}{2}$ **B** 10 **C** -10 **D** $\dfrac{3}{2}$

d Describe the behaviour of the gradient of the tangent as x decreases.

12 Find the gradient of $f(x) = (x-5)^3$ at the [redacted]t $(8, 27)$.

13 Find the gradient of the tangent to the curve $y = (3x-1)^3$ at the point $(1, 8)$.

14 Find $f'(2)$ for the function $f(x) = 2\left(x^2 - 7\right)^5$.

15 Find the x-coordinate of the point at which $f(x) = (x-2)^2$ has a gradient of 6.

16 Find the x-coordinate(s) of the point(s) at which $f(x) = (x+2)^3$ has a gradient of 48.

17 Find the values of x where the tangent of $y = \left(x^2 - 1\right)^3$ is horizontal.

18 Find the values of x where the derivative of $y = \left(2x + x^2\right)^5$ is equal to zero.

19 The curve $y = \sqrt{x-3}$ has a tangent with a gradient of $\frac{1}{2}$ at the point P. Find the coordinates of P.

20 For the function $f(x) = \dfrac{1}{2x-7}$, find the values of x where $f'(x) = -\dfrac{2}{25}$.

21 Consider the semicircle defined as $y = \sqrt{169 - x^2}$.

a Find the derivative of the function.

b Sketch the graph of the semicircle.

c Find the equation of the tangent at the point $(5, 12)$.

22 Find the equation of the tangent to $y = (2x+1)^4$ at the point where $x = -1$.

23 Find the equation of the tangent to $y = (2x-1)^8$ at the point where $x = 1$.

24 Find the equation of the normal to $y = (3x-4)^3$ at the point $(1, -1)$.

25 Find the equation of the normal to $y = \left(x^2 + 1\right)^4$ at the point where $x = 1$.

26 Consider the curve $f(x) = \dfrac{1}{2x+3}$ at the point $(-1, 1)$.

a Find the equation of the tangent at the point.

b Find the equation of the normal at the point.